OBC-V2G22

PRODUCT REQUIREMENTS DOCUMENT (PRD)

800 V Bi-directional 22 kW On-Board Charger with Vehicle-to-Grid

Automotive EV Power Electronics | Rev A1 - Architecture, Atomic Requirements, Interfaces & Verification Baseline

Mission profile. The product converts grid AC power to an 800 V-class traction battery for charging and returns battery energy to the grid in approved V2G operating modes. The source baseline fixes a 22 kW bidirectional architecture using a totem-pole PFC, isolated dual-active bridge, 1200 V SiC MOSFETs, AUTOSAR Classic Platform control, model-predictive ZVS optimization, ASIL-C safety intent, liquid cooling, CAN FD/SecOC, IP6K9K packaging, automated HiPot and cryptographic EOL provisioning.

Engineering baseline. This document preserves the uploaded portfolio statements as the source-defined product baseline, then expands them into an engineering review baseline using atomic requirements, controlled parameters, interface ownership, safety mechanisms and objective verification evidence. Values not supplied by the source are treated as TBC, variant-controlled, supplier-dependent or installation-dependent rather than silently promoted to released limits.

Definition	Controlled product interpretation
Product class	22 kW bidirectional on-board charger for an 800 V-class EV traction system with grid-interactive V2G capability.
Safety / assurance	Source baseline: ASIL-C compliant with hardware-based overcurrent lockouts; ISO/SAE 21434 cybersecurity; program-specific ISO 26262 allocation to be confirmed.
Lifecycle mission	15-year vehicle lifecycle with NVM wear-leveling, thermal-cycle logging, EOL configuration control and evidence-based change impact.
Primary architecture	AC input protection/EMI filter -> bidirectional totem-pole PFC -> DC link -> isolated DAB -> HV battery; 1200 V SiC; liquid cold plate; AUTOSAR CP supervisory/control software.
Release philosophy	Hardware-independent product limits are separated from installation, supplier, calibration and vehicle-variant values. Every atomic requirement traces to objective verification evidence.

Source basis: 15_Future_Proof_PRDs_Engineering_Expansion_Review_Baseline(1).pdf, Document 1. Expansion style benchmark: EMA_FC540_Flight_Critical_270V_540V_Actuator_Inverter_PRD_RevA3_Parchment_Cleaned (1).pdf. AI-assisted engineering expansion requires responsible engineer review before release.

PRODUCT GOVERNANCE, REQUIREMENT QUALITY & CHANGE CONTROL

Single-intent requirements, controlled parameters, explicit interface ownership and evidence-based release.

Safety mechanisms are allocated by physical response time and independence. Fast destructive faults such as semiconductor overcurrent, shoot-through or isolation-bias collapse are contained by autonomous hardware paths where application-software latency is not credible. Supervisory software owns classification, logging, degraded operation, service reporting and controlled recovery. Network commands shall never bypass local electrical, thermal or insulation limits.

Requirements in this PRD are intentionally atomic. A requirement is considered closed only when the released hardware revision, software build, calibration set, test equipment, environmental condition and vehicle/grid simulator configuration are identified in the evidence package. Planning targets and supplier typical values are not treated as production guarantees.

Parameter class	Examples / release rule
Immutable safety	Absolute current/voltage/temperature limits, hard gate inhibit, independent trip paths. Change requires safety impact analysis and affected DVP regression.
Variant controlled	AC region, connector type, battery range, cooling interface, power rating, software feature set. Released by signed variant code, BOM and calibration package.
Supplier dependent	SiC SOA/short-circuit withstand, isolated driver CMTI, capacitor life, magnetics insulation, sensor error. Use guaranteed production data and worst-case analysis.
Installation dependent	Grid impedance, PE/grounding, harness length, coolant conditions, packaging heat rejection and EMC environment. Captured in vehicle/charger ICD and integration DVP.
Calibration	Loop gains, thresholds, dead time, derate curves, timing, filters and mode maps. Authenticated, range checked and traceable to verification evidence.

Traceability rule. Each product requirement shall trace upward to a product mission, hazard/risk control or external interface obligation and downward to at least one verification artifact. Evidence reuse after a semiconductor, topology, software, protocol or installation change is permitted only after documented similarity analysis proves the changed parameter remains inside the assumptions of the linked analyses; otherwise the impacted test, model and safety argument are reopened.

SYSTEM BOUNDARY, ASSUMPTIONS & EXTERNAL OWNERSHIP

The OBC is one element of the vehicle charging function. It receives AC grid energy and vehicle/battery commands and exchanges high-voltage DC energy with the traction battery. Grid service disconnects, EVSE protection, battery contactors, coolant supply and vehicle supervisory permission remain external unless allocated by the vehicle ICD.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-SYS-001	The released OBC boundary shall identify ownership of AC protection, EVSE interlock, protective earth, HV battery contactors, coolant supply, wake/sleep authority and communication permissions.	System architecture and ICD review.
OBC-SYS-002	External mitigations credited in the safety case shall have explicit response-time, diagnostic and failure-independence assumptions.	Safety/ICD trace review.
OBC-SYS-003	Loss of an external permissive required for power transfer shall transition the OBC to the allocated non-energizing or controlled-energy state without increasing grid or battery current.	HIL/power-HIL permissive-loss test.
OBC-SYS-004	The source-defined 22 kW rating shall not be interpreted as a universal charge-time guarantee independent of battery usable energy, SOC window, grid limit and thermal conditions.	Product requirement review.

Verification, margin and future-change rule

Changes to EVSE architecture, battery contactor ownership, vehicle wake strategy or cooling boundary reopen the interface safety and sequencing analysis.

OPERATING MODES & STATE AUTHORITY

The product supports charge, V2G discharge, standby, sleep, precharge, diagnostic/service, controlled shutdown and fault-latched modes. Mode transitions are permission-based; power conversion is enabled only after sensing, communication validity, isolation/ground checks and local protection readiness meet the released criteria.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-MOD-010	Each released mode shall define entry conditions, exit conditions, command authority, power limits, timeout behavior and safe-state response.	State-machine review and HIL transition matrix.
OBC-MOD-011	A transition between charge and V2G shall pass through a controlled zero/limited-power state unless the released control analysis proves bumpless bidirectional reversal.	Power-HIL reversal test.
OBC-MOD-012	Reset, brownout or watchdog recovery shall not create an uncontrolled gate-enable or power-flow step.	Power-cycle and reset fault injection.
OBC-MOD-013	Conflicting vehicle, EVSE or grid-service commands shall resolve to a deterministic non-permissive authority rule.	Command-conflict HIL test.

Verification, margin and future-change rule

Any new operating mode, bidirectional transition policy or command owner requires state-machine regression and safety impact review.

AC INPUT, CONTACTOR INTERFACE & INRUSH CONTROL

The AC boundary includes connector/input pins, fusing/protection assumptions, line sensing, EMI filter, discharge paths and any internal relay/contactor functions. Normal, abnormal and transient input envelopes are variant-controlled because grid region and EVSE architecture differ.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-IN-020	Each AC variant shall define normal, abnormal and transient voltage/frequency envelopes used for component, insulation and control sizing.	Electrical ICD and bench sweep.
OBC-IN-021	Gate enable shall remain inhibited until AC sensing, DC-link state, control bias and protection health meet READY criteria.	Power sequencing test.
OBC-IN-022	Input inrush and DC-link energization shall remain within connector, fuse, relay and semiconductor stress limits for the released source impedance.	Inrush/precharge test with worst source impedance.
OBC-IN-023	Loss or bounce of an input relay/contactor signal shall not produce repeated destructive charging of the DC link.	Relay-bounce fault injection.

Verification, margin and future-change rule

AC-region, connector, EVSE or feeder-impedance changes reopen inrush, transient and EMC verification.

BIDIRECTIONAL TOTEM-POLE PFC POWER STAGE

The source architecture uses a totem-pole PFC with 1200 V SiC MOSFETs. Device selection is based on worst bus/transient voltage, switching overshoot, reverse conduction, short-circuit/fault energy and hot-junction margin rather than nominal grid voltage alone.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-PFC-030	PFC power devices shall retain released voltage, current, temperature and transient SOA at maximum AC/DC stress and commutation overshoot.	WCCA plus double-pulse/full-power switching test.
OBC-PFC-031	Commutation-loop inductance and gate profile shall meet the overshoot used in device derating analysis.	DPT and layout correlation.
OBC-PFC-032	Dead-time and reverse-conduction behavior shall prevent cross-conduction across the released voltage, temperature and tolerance envelope.	Worst-corner switching test.
OBC-PFC-033	A single switch/driver fault shall transition the stage to the allocated safe response before energy exceeds the protected component capability.	Fault-injection/short-circuit test.

Verification, margin and future-change rule

SiC device, package, gate driver or power-loop layout changes trigger switching, EMI, thermal and protection regression.

DC LINK, PRECHARGE, DISCHARGE & STORED ENERGY

The DC-link architecture stores enough energy to create hazardous stress after grid removal. Capacitance, ESR/ESL, bleed/discharge, precharge/inrush and voltage sensing therefore form a single controlled energy-management function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-DCL-040	DC-link capacitance and ripple-current capability shall be sized for the released charge/V2G operating envelope and end-of-life loss budget.	WCCA, thermal and ripple-current validation.
OBC-DCL-041	The DC link shall not be declared safe-to-service until measured voltage is below the released threshold and any diagnostic uncertainty is accounted for.	Discharge timing and sensor-fault test.
OBC-DCL-042	Precharge/discharge elements shall remain within repetitive pulse-energy and temperature limits under restart and fault-recovery sequences.	Repeated-cycle energy test.
OBC-DCL-043	Invalid DC-link voltage data shall force a non-permissive state for gate enable and service discharge completion.	Frozen/stuck sensor injection.

Verification, margin and future-change rule

Capacitor technology, capacitance, precharge topology or service-discharge requirement changes reopen stored-energy and lifetime evidence.

ISOLATED DUAL-ACTIVE-BRIDGE DC/DC

The DAB provides galvanic isolation and bidirectional DC/DC conversion between the regulated DC link and the HV battery. Phase-shift modulation, transformer ratio, leakage inductance and device timing jointly determine power range, circulating current and ZVS behavior.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-DAB-050	The DAB shall meet the released bidirectional power envelope without exceeding semiconductor, transformer, inductor or capacitor current/voltage limits.	Full-power dyno/power-HIL sweep.
OBC-DAB-051	Phase-shift and dead-time commands shall be bounded to prevent loss of controllability or transformer flux imbalance.	SIL/HIL plus switching waveform validation.
OBC-DAB-052	Loss of ZVS shall be detected or bounded by a derated operating region such that device dissipation remains within thermal limits.	ZVS map correlation across corners.
OBC-DAB-053	A bridge-leg fault shall not transfer uncontrolled energy across the isolation barrier beyond the allocated protection response.	Fault containment test/analysis.

Verification, margin and future-change rule

Transformer ratio, leakage inductance, modulation strategy or device technology changes require power-map, ZVS, thermal and insulation regression.

ISOLATION TRANSFORMER, MAGNETICS & HIGH-FREQUENCY INSULATION

The transformer and filter magnetics are safety- and performance-critical because insulation stress, core loss, copper loss, parasitic capacitance and leakage inductance influence efficiency, common-mode current and DAB commutation. Manufacturing variation is included in the released model.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-MAG-060	The isolation transformer shall meet released working/transient voltage, creepage/clearance, insulation-system and partial-discharge requirements across lifetime and environment.	HiPot/IR plus PD characterization where applicable.
OBC-MAG-061	Core flux density and copper/core hotspot temperature shall remain within released limits for maximum power, frequency and modulation corners.	Thermal mapping and magnetic WCCA.
OBC-MAG-062	Leakage and magnetizing inductance production variation shall remain within the control/ZVS assumptions.	EOL Lm/Llk measurement and correlation.
OBC-MAG-063	Parasitic capacitance/common-mode current shall remain within the EMC/leakage-current architecture.	Impedance extraction and conducted-emission/leakage test.

Verification, margin and future-change rule

Magnetic material, winding process, insulation system or supplier change triggers electrical, thermal, PD and EMC requalification.

ISOLATED GATE DRIVE, FAST OVERCURRENT & ACTIVE TURN-OFF

Destructive power-stage faults require autonomous gate-off independent of the AUTOSAR task. Gate drivers provide UVLO, CMTI margin, desaturation/overcurrent detection where appropriate, controlled turn-off and latched fault reporting.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-GD-070	Fault-onset-to-safe-current time shall remain within the selected power-device minimum guaranteed fault-withstand allocation.	Worst Vdc/Tj short-circuit test.
OBC-GD-071	Fast gate inhibit shall operate independently of the application software and network stack.	Processor-stall and bus-overload test.
OBC-GD-072	Active clamp or multi-stage turn-off shall keep device voltage and fault energy inside released SOA.	Fault waveform/energy integration.
OBC-GD-073	CMTI and off-state gate margin shall prevent false turn-on at worst qualified dv/dt.	DPT/CMTI susceptibility test.

Verification, margin and future-change rule

Gate-driver, isolated-bias, turn-off network, SiC package or layout changes trigger fast-fault and dv/dt regression.

VOLTAGE, CURRENT & TEMPERATURE METROLOGY

Accurate sensing supports grid current control, battery current control, power calculation, protection, diagnostics and thermal derating. Safety mechanisms do not rely on a single unqualified conversion chain when an independent fast path is required.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-SNS-080	Each safety- or control-relevant sensor channel shall define range, accuracy, bandwidth, offset/gain drift, plausibility and diagnostic behavior over environment and lifetime.	Sensor WCCA and calibrated bench sweep.
OBC-SNS-081	Current/voltage measurement latency shall remain within control stability and protection timing allocations.	Timing correlation and step-response test.
OBC-SNS-082	Stale, frozen, saturated or implausible measurements shall propagate an invalid status rather than being silently reused.	Fault-injection test.
OBC-SNS-083	Thermal estimation shall remain conservative when one temperature sensor is unavailable or implausible.	Thermal sensor dropout test.

Verification, margin and future-change rule

Sensor topology, ADC reference, isolation amplifier or filtering changes require metrology, timing and safety revalidation.

CONTROL ARCHITECTURE, PFC LOOPS & DAB POWER CONTROL

The software baseline uses AUTOSAR Classic Platform with model-predictive control for ZVS optimization. Fast current loops, grid synchronization, DC-link control, DAB power control and thermal limits are partitioned so supervisory features cannot override local protection.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-CTL-090	Control-loop sample/PWM timing shall be deterministic and synchronized to the released switching strategy.	Timing trace and SIL/HIL verification.
OBC-CTL-091	PFC current and DC-link loops shall meet released stability margins over grid, component and thermal tolerances.	Frequency-response and power-HIL sweep.
OBC-CTL-092	DAB power control shall bound phase shift, current and power commands across battery voltage and bidirectional flow.	Power-HIL corner matrix.
OBC-CTL-093	Control saturation and recovery shall use bounded rate/anti-windup behavior without creating a power step.	Saturation/recovery test.

Verification, margin and future-change rule

Switching frequency, control rate, loop ownership or plant parameter changes require stability and timing regression.

ZVS OPTIMIZATION & MODEL-PREDICTIVE CONTROL

Model-predictive ZVS optimization is treated as an efficiency/performance layer constrained by hard current, voltage and thermal limits. Invalid model inputs or prediction failure shall reduce capability rather than widen protection limits.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-ZVS-100	The released ZVS optimization shall define valid operating region, model inputs, update rate and fallback behavior.	Model review and HIL boundary sweep.
OBC-ZVS-101	Prediction/model error shall not command phase shift or current outside independently enforced limits.	Model-mismatch fault injection.
OBC-ZVS-102	Loss of ZVS optimization shall transition to a bounded fallback modulation strategy with verified thermal capability.	Fallback efficiency/thermal test.
OBC-ZVS-103	Calibration updates affecting ZVS maps shall be authenticated, range checked and traceable to verification evidence.	Calibration-security and regression audit.

Verification, margin and future-change rule

Control-model, calibration or device-loss model changes trigger ZVS-map and thermal regression before release.

GRID SYNCHRONIZATION, POWER QUALITY & V2G EXPORT

Bidirectional operation requires valid grid synchronization and controlled real/reactive power behavior. Exact regional grid-code limits are installation/program parameters and remain TBC until the target market and certification plan are fixed.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-GRID-110	V2G power export shall be enabled only when grid voltage, frequency, phase validity and external permission meet the released regional criteria.	Grid-emulator conformance test.
OBC-GRID-111	Loss of grid synchronization or islanding permissive shall command a controlled non-exporting state within the allocated response.	Grid-disturbance/islanding test.
OBC-GRID-112	Charge and export current harmonics, power factor and DC injection shall meet the released market-specific limits.	Power-quality test with calibrated analyzer.
OBC-GRID-113	Reactive-power or grid-service commands shall remain subordinate to OBC thermal, current, battery and safety limits.	Command-priority power-HIL test.

Verification, margin and future-change rule

Grid-code, EVSE/DER profile or market change requires regional parameter release and power-quality regression.

BATTERY HV INTERFACE, CONTACTOR COORDINATION & REGEN ENERGY

The OBC does not unilaterally energize the battery bus. Battery contactor/precharge authority, pack voltage/current limits, insulation state and BMS permission are explicit interfaces. During V2G, battery discharge limits remain authoritative.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-BAT-120	HV output enable shall require valid battery-voltage range, BMS permission, contactor state and insulation/safety readiness.	Vehicle-HIL/power-HIL sequence test.
OBC-BAT-121	OBC current command shall remain within the freshest valid BMS charge/discharge limits and local hardware limits.	Limit-step/freshness fault injection.
OBC-BAT-122	Loss of BMS communication shall transition to the released non-permissive current state without an uncontrolled DC-link excursion.	Communication-loss power test.
OBC-BAT-123	Contactor mismatch or unexpected HV presence shall block gate enable and raise a latched diagnostic according to the safety concept.	Contactor fault injection.

Verification, margin and future-change rule

Battery voltage class, BMS protocol, contactor ownership or pack precharge strategy changes reopen the HV interface and safety sequence.

FUNCTIONAL SAFETY, FTTI & INDEPENDENT PROTECTION

The source states ASIL-C compliance with hardware-based overcurrent lockouts. Detailed HARA/FSC/TSC allocation remains program-specific. This PRD therefore defines safety mechanisms and timing ownership without inventing vehicle-level hazard classifications.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-SAF-130	Each safety goal allocated to the OBC shall trace to one or more item requirements, mechanisms, diagnostics and verification evidence.	Safety traceability audit.
OBC-SAF-131	Fast overcurrent and shoot-through protection shall not depend on successful execution of the AUTOSAR application task.	Processor-stall plus fault test.
OBC-SAF-132	Safe/degraded states shall define whether AC, DC and gate paths are inhibited, discharged or limited after each detected high-severity fault.	Fault-response matrix test.
OBC-SAF-133	Fault detection plus reaction time shall remain inside the allocated FTTI/protected-component withstand budget at worst environment.	Timed fault-injection evidence.

Verification, margin and future-change rule

Any safety-goal, protection path, ASIL allocation or external mitigation change requires ISO 26262 impact analysis and affected DVP regression.

LIQUID COOLING, HOT-SOAK RESTART & DERATING

The source requires a liquid-cooled cold plate and Tj below 150 C at full 22 kW load. That source value is preserved as a top-level ceiling; the released design shall use lower component-specific operating/derating limits where required by guaranteed lifetime data.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-THM-140	The cooling interface shall define coolant inlet temperature, flow/pressure boundary, cold-plate resistance and allowable degradation as controlled parameters.	Thermal ICD and bench characterization.
OBC-THM-141	Estimated semiconductor junction temperature shall remain below the source ceiling of 150 C and below any lower released device/lifetime limit at full 22 kW operation.	Thermal mapping and loss-model correlation.
OBC-THM-142	Hot-soak restart shall apply verified derating or inhibit when thermal margin is insufficient for requested power.	Hot-soak restart test.
OBC-THM-143	Loss or invalidity of coolant/temperature data shall reduce capability or shut down without increasing thermal stress.	Cooling-fault and sensor-fault test.

Verification, margin and future-change rule

Cold plate, TIM, coolant, power device or enclosure changes trigger thermal model correlation and degraded-cooling regression.

CYBERSECURITY, CAN FD / SECOC & COMMAND AUTHORITY

The source specifies ISO/SAE 21434 compliance and CAN FD with SecOC. The security design protects software, calibration, diagnostics, V2G permissions and provisioning while maintaining independence of immediate local hardware protections.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-CYB-150	Safety- or power-authority CAN FD messages shall define source, destination, freshness, integrity, timeout and non-permissive fallback.	SecOC/protocol negative testing.
OBC-CYB-151	Only authenticated approved software and calibration shall execute in a power-enabled state.	Modified-image/load negative test.
OBC-CYB-152	Loss or invalidity of cybersecurity services shall reject untrusted commands without delaying local hardware gate-off or thermal protection.	Concurrent cyber-fault/safety test.
OBC-CYB-153	Production key injection and certificate identity shall be traceable to the serialized hardware unit.	EOL provisioning audit.

Verification, margin and future-change rule

Cryptographic algorithm, key-management backend, CAN database or bootloader changes trigger TARA and security regression.

MECHANICAL PACKAGING, IP6K9K & HV CONNECTORS

The source specifies an IP6K9K die-cast housing with vibration-resistant AC/DC headers. The enclosure must simultaneously support sealing, HV creepage/clearance, thermal conduction, pressure equalization, EMC bonding and serviceability.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-MEC-160	The released enclosure shall maintain IP6K9K sealing and connector retention after the defined vibration, thermal-cycle and pressure-wash sequence.	Sequential environmental/ingress test.
OBC-MEC-161	HV/AC connector keying and retention shall prevent incompatible mating and maintain required creepage/clearance after tolerance stack-up.	Mechanical/insulation inspection and test.
OBC-MEC-162	Enclosure bonding and gasket/fastener strategy shall support the EMC and protective-earth architecture throughout lifecycle wear.	Bonding resistance and EMC correlation.
OBC-MEC-163	Service disassembly/reassembly shall not silently invalidate sealing, thermal interface or HV safety without a controlled replacement process.	Serviceability/process validation.

Verification, margin and future-change rule

Housing, connector, vent, gasket, fastener or mounting changes require mechanical, ingress, EMC and thermal delta analysis.

MANUFACTURING, EOL HIPOT & CRYPTOGRAPHIC PROVISIONING

The source calls for automated HiPot and end-of-line cryptographic key injection. Production release therefore couples electrical safety test, calibration, traceability, variant programming and security identity to the serialized unit.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-MFG-170	EOL test shall verify safety-critical power, sensing, gate-inhibit, communication and discharge functions with objective pass/fail limits.	EOL fixture qualification and golden-unit correlation.
OBC-MFG-171	Automated HiPot/insulation tests shall use configuration-controlled voltage, dwell, leakage limit and measurement uncertainty.	HiPot gauge capability and audit.
OBC-MFG-172	Cryptographic key/certificate injection shall be secure, serialized and protected against duplicate or unauthorized provisioning.	Provisioning process security audit.
OBC-MFG-173	Software trim shall not be used to mask an assembly defect that consumes safety or performance margin.	Process capability and calibration review.

Verification, margin and future-change rule

Fixture, test method, calibration algorithm, provisioning station or critical supplier process changes require manufacturing impact review and correlation.

15-YEAR LIFECYCLE, AGING COUNTERS & FIELD DATA

The source specifies NVM wear-leveling and thermal-cycle logging for warranty tracking. Lifecycle evidence also covers power-device cycling, capacitor aging, magnetic insulation, sealing, connector wear, coolant interface degradation and software/configuration obsolescence.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBC-LIFE-180	The released mission profile shall define energy throughput, thermal cycles, charge/V2G duty, environment and service events used for lifetime design.	Mission-profile review and reliability plan.
OBC-LIFE-181	NVM shall use wear management and retain critical event/cycle data over the released lifecycle with corruption recovery.	Endurance and power-interruption testing.
OBC-LIFE-182	Thermal-cycle and high-stress event counters shall be configuration-controlled and traceable to field diagnostics without being treated as sole proof of remaining life.	Field-data and diagnostic audit.
OBC-LIFE-183	Supplier or component obsolescence substitutions shall require similarity analysis against electrical, thermal, EMC, safety and lifetime assumptions.	Change-control review.

Verification, margin and future-change rule

Any mission-profile expansion or component substitution beyond qualified similarity bounds triggers targeted reliability and DVP regression.

PART II - DETAILED SUBSYSTEM PRDS & ENGINEERING ASSURANCE ANNEX

Expanded architecture, atomic requirements, fault containment, interfaces, DVP and lifecycle evidence.

This annex extends the product-level OBC baseline into subsystem requirements for design review, hardware/software allocation, ISO 26262 work products, cybersecurity analysis, interface definition, production release and lifecycle change control. Controlled parameters are used where the final vehicle, grid region, EVSE, battery, harness, coolant, software load or supplier configuration has not fixed a value.

Across Part II, every TBC shall have an owner and closure method. Evidence shall identify the released configuration, calibrated instrumentation and raw records needed to demonstrate the mechanism. Prior evidence may be reused only after documented similarity and impact review shows the affected stress and margin remain bounded; otherwise the impacted analysis and DVP shall be repeated.

Annex family	Primary closure intent
A01-A05	Power-on state, AC boundary, input transient, DC-link energy, PFC metrology
A06-A10	SiC switching, DAB energy transfer, magnetics, insulation, gate-drive protection
A11-A15	Timing, controls, ZVS, thermal, leakage/ground-fault
A16-A20	EMC, communications, V2G authority, diagnostics, cybersecurity
A21-A25	AUTOSAR, boot/update, calibration, manufacturing, EOL
A26-A30	Full-envelope DVP, environmental, lifetime, supplier change, release evidence

AC / VEHICLE POWER-ON STATE MACHINE

Detailed subsystem PRD / engineering assurance annex - control closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA01-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for ac / vehicle power-on state machine; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA01-02	Controller states, gate-enable conditions, precharge/discharge ownership, timeouts and invalid-input behavior shall be bounded across the released source/load and sensor-tolerance envelope.	SIL/HIL or power-HIL state transition sweep with timing, brownout and invalid-sensor injection.
OBCA01-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Incorrect sequencing can connect energy before line, DC-link, isolation, battery contactor or controller readiness is qualified. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is sil/hil or power-hil state transition sweep with timing, brownout and invalid-sensor injection. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

AC SURGE, DIP, INTERRUPT & RESTART BOUNDARY

Detailed subsystem PRD / engineering assurance annex - electrical closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA02-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for ac surge, dip, interrupt & restart boundary; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA02-02	Normal, abnormal and transient AC conditions shall map to deterministic continue/derate/stop/restart behavior without exceeding stored-energy or device stress limits.	Programmable AC-source sweep with surge/dip/interruption and restart logging.
OBCA02-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Line disturbances can trigger repeated restart, uncontrolled DC-link energy or nuisance protection unless restart hysteresis and recovery are explicit. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is programmable ac-source sweep with surge/dip/interruption and restart logging. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

INPUT EMI FILTER DAMPING & GRID IMPEDANCE STABILITY

Detailed subsystem PRD / engineering assurance annex - hardware closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA03-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for input emi filter damping & grid impedance stability; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA03-02	Input filter damping and control shall remain stable across the released grid/source impedance and component tolerance envelope.	Impedance analysis correlated to network analyzer and power-HIL weak-grid tests.
OBCA03-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that The EMI filter and active PFC input impedance can interact with weak grids or EVSE impedance and create oscillation even when nominal conducted emissions pass. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is impedance analysis correlated to network analyzer and power-hil weak-grid tests. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

PFC COMMUTATION LOOP, DEAD TIME & REVERSE CONDUCTION

Detailed subsystem PRD / engineering assurance annex - power-stage closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA04-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for pfc commutation loop, dead time & reverse conduction; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA04-02	Gate timing, loop inductance and reverse-conduction interval shall stay within device SOA and released loss/EMI limits over tolerance and temperature.	Double-pulse matrix with timing/temperature/layout correlation.
OBCA04-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Fast SiC switching can trade efficiency for overshoot, EMI and false turn-on. Dead time must remain safe over propagation mismatch and temperature. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is double-pulse matrix with timing/temperature/layout correlation. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

PFC CURRENT METROLOGY & GRID CURRENT ACCURACY

Detailed subsystem PRD / engineering assurance annex - metrology closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA05-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for pfc current metrology & grid current accuracy; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA05-02	Current measurement shall retain released accuracy, linearity and latency from minimum current through full bidirectional power and during commutation common-mode events.	Calibrated source/load sweep and dynamic step test.
OBCA05-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Current-sense error directly affects power quality, protection margin and exported V2G current. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is calibrated source/load sweep and dynamic step test. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

DC-LINK CAPACITOR RIPPLE, ESR & END-OF-LIFE MARGIN

Detailed subsystem PRD / engineering assurance annex - lifetime closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA06-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for dc-link capacitor ripple, esr & end-of-life margin; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA06-02	Capacitance, ESR and ripple-current margins shall include temperature, tolerance, mission profile and released end-of-life degradation.	WCCA, accelerated aging correlation and full-power ripple measurement.
OBCA06-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Capacitor ESR growth and capacitance loss can increase ripple, heating and control stress late in life. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is wcca, accelerated aging correlation and full-power ripple measurement. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

DAB PHASE-SHIFT LIMITS & CIRCULATING CURRENT

Detailed subsystem PRD / engineering assurance annex - control closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA07-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for dab phase-shift limits & circulating current; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA07-02	DAB modulation shall enforce bounded phase shift/current for every released battery/DC-link ratio and direction of power flow.	Power-HIL/full-power corner sweep with current waveform capture.
OBCA07-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that A mathematically valid phase-shift command can still create excessive circulating current or transformer stress at light load or extreme voltage ratio. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is power-hil/full-power corner sweep with current waveform capture. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

TRANSFORMER FLUX BALANCE & SATURATION PROTECTION

Detailed subsystem PRD / engineering assurance annex - magnetics closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA08-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for transformer flux balance & saturation protection; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA08-02	The released design shall prevent or detect flux imbalance before core saturation or winding current exceeds protected limits.	Injected asymmetry test plus magnetic model correlation.
OBCA08-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Control asymmetry, sensor offset or faulted gating can accumulate volt-seconds and saturate the transformer. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is injected asymmetry test plus magnetic model correlation. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

HIGH-FREQUENCY INSULATION & PARTIAL-DISCHARGE MARGIN

Detailed subsystem PRD / engineering assurance annex - safety closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA09-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for high-frequency insulation & partial-discharge margin; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA09-02	Working voltage, repetitive transient crest, material system, creepage/clearance, PD margin and manufacturing variation shall be configuration-controlled.	HiPot/IR plus PDIV/PDEV or repetitive-impulse characterization where applicable.
OBCA09-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Repetitive SiC dv/dt can stress insulation differently from DC HiPot and may create partial-discharge aging without immediate functional symptoms. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is hipot/ir plus pdiv/pdev or repetitive-impulse characterization where applicable. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

ISOLATED BIAS SUPPLY UVLO & COMMON-CAUSE FAULTS

Detailed subsystem PRD / engineering assurance annex - safety closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA10-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for isolated bias supply uvlo & common-cause faults; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA10-02	Isolated bias architecture shall default gates OFF on undervoltage and contain a single bias fault within the allocated protection independence.	Bias brownout, short/open and common-cause fault injection.
OBCA10-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that A gate-driver bias fault can force a device into linear operation or defeat both directions of a bridge if common supplies are not contained. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is bias brownout, short/open and common-cause fault injection. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

AUTOSAR ISR TIMING, PWM SYNCHRONIZATION & WCET

Detailed subsystem PRD / engineering assurance annex - software closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA11-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for autosar isr timing, pwm synchronization & wcet; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA11-02	Measured WCET, interrupt latency and PWM/ADC synchronization shall retain release-controlled timing reserve under maximum communication, diagnostic and fault-interrupt load.	Instrumented WCET stress test and HIL timing injection.
OBCA11-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Timing jitter or overload can destabilize current loops or delay safety classification even when average CPU utilization is acceptable. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is instrumented wcet stress test and hil timing injection. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

PFC LOOP STABILITY ACROSS WEAK GRID & DC-LINK CORNERS

Detailed subsystem PRD / engineering assurance annex - control closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA12-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for pfc loop stability across weak grid & dc-link corners; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA12-02	Loop gain/phase margins shall remain within released stability criteria for all approved grid/source and power-flow corners.	Frequency-response identification and power-HIL sweep.
OBCA12-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Outer DC-link and inner current loops must remain stable across source impedance, line frequency, DC-link loading and parameter tolerance. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is frequency-response identification and power-hil sweep. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

MODEL-PREDICTIVE ZVS ESTIMATION ERROR & FALLBACK

Detailed subsystem PRD / engineering assurance annex - control closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA13-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for model-predictive zvs estimation error & fallback; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA13-02	The MPC layer shall be bounded by independent electrical/thermal limits and enter a validated fallback when model confidence or required inputs are invalid.	Model mismatch, stale-input and fallback thermal tests.
OBCA13-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Model error can move the commanded operating point out of ZVS or into higher current/loss regions. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is model mismatch, stale-input and fallback thermal tests. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

COLD-PLATE THERMAL RESISTANCE & HOT-SPOT CORRELATION

Detailed subsystem PRD / engineering assurance annex - thermal closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA14-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for cold-plate thermal resistance & hot-spot correlation; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA14-02	Junction, magnetics and capacitor hot spots shall remain below released end-of-life limits for the full duty cycle including degraded cooling/interface resistance.	Thermal mapping and faulted-cooling tests correlated to an end-of-life model.
OBCA14-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Beginning-of-life thermal margin can disappear with TIM pump-out, coolant degradation or assembly variation. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is thermal mapping and faulted-cooling tests correlated to an end-of-life model. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

LEAKAGE CURRENT, PE / GROUND-FAULT & RESIDUAL CURRENT

Detailed subsystem PRD / engineering assurance annex - safety closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA15-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for leakage current, pe / ground-fault & residual current; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA15-02	The released design shall define PE/bonding, common-mode capacitance, leakage/residual-current behavior and response to insulation/ground faults for the vehicle/EVSE architecture.	Leakage/residual-current and insulation-fault injection testing.
OBCA15-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that High-frequency common-mode current can affect touch/leakage behavior and external residual-current detection. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is leakage/residual-current and insulation-fault injection testing. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

CONDUCTED / RADIATED EMC & CABLE COMMON-MODE CURRENT

Detailed subsystem PRD / engineering assurance annex - EMC closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA16-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for conducted / radiated emc & cable common-mode current; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA16-02	EMI filters, shielding, bonding and edge-rate control shall meet released emissions/susceptibility limits without consuming protection or thermal margin.	CISPR/vehicle-level EMC test plus current-probe correlation.
OBCA16-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that SiC edge rates couple through AC, HV, coolant/structure and communication paths, so local switch-node optimization alone is insufficient. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is cispr/vehicle-level emc test plus current-probe correlation. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

CAN FD SIGNAL OWNERSHIP, FRESHNESS & TIMEOUT

Detailed subsystem PRD / engineering assurance annex - interface closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA17-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for can fd signal ownership, freshness & timeout; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA17-02	Every safety- or performance-relevant CAN FD signal shall define source, units/range, update rate, SecOC/freshness rule, timeout and non-permissive fallback.	Protocol conformance with stale/range/integrity fault injection.
OBCA17-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Stale or ambiguously owned battery/grid permission can be more dangerous than complete communication loss. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is protocol conformance with stale/range/integrity fault injection. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

V2G AUTHORIZATION & CHARGE / EXPORT REVERSAL

Detailed subsystem PRD / engineering assurance annex - system closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA18-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for v2g authorization & charge / export reversal; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA18-02	Charge-to-export and export-to-charge transitions shall use a configuration-controlled authorization sequence and bounded power slew.	Grid/battery power-HIL reversal matrix.
OBCA18-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that A reversal in energy direction must coordinate grid authorization, BMS discharge permission, DC-link control and thermal capability. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is grid/battery power-hil reversal matrix. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

FAULT CLASSIFICATION, LATCHING & RE-ARM POLICY

Detailed subsystem PRD / engineering assurance annex - diagnostic closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA19-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for fault classification, latching & re-arm policy; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA19-02	Each diagnostic shall define severity, debounce, latch behavior, power-limit effect, service data and explicit re-arm authority.	Fault-injection and repeated-recovery test.
OBCA19-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Automatic re-arm can convert an intermittent short, insulation fault or thermal problem into repeated destructive cycling. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is fault-injection and repeated-recovery test. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

SECURE BOOT, SIGNED UPDATE & CALIBRATION PROTECTION

Detailed subsystem PRD / engineering assurance annex - cyber closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA20-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for secure boot, signed update & calibration protection; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA20-02	Only authenticated approved software/calibration shall execute in a power-enabled state; recovery shall preserve local hardware protections.	Modified-image, rollback and corrupted-calibration negative tests.
OBCA20-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Compromised software or calibration could change power limits, dead time, protection thresholds or V2G authority. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is modified-image, rollback and corrupted-calibration negative tests. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

AUTOSAR PARTITIONING, WATCHDOG & FREEDOM FROM INTERFERENCE

Detailed subsystem PRD / engineering assurance annex - software closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA21-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for autosar partitioning, watchdog & freedom from interference; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA21-02	Task/resource allocation, watchdog servicing and memory/CPU protection shall preserve control timing and fault response under worst defined load.	WCET, overload and watchdog fault tests.
OBCA21-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Diagnostics, communication or logging bursts must not starve fast control or safety supervision. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is wcet, overload and watchdog fault tests. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

NVM WEAR-LEVELING, EVENT LOGGING & POWER-FAIL INTEGRITY

Detailed subsystem PRD / engineering assurance annex - lifecycle closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA22-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for nvm wear-leveling, event logging & power-fail integrity; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA22-02	NVM architecture shall bound write endurance, preserve critical first-fault data and recover consistently from interrupted writes.	Accelerated endurance and power-cut test.
OBCA22-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Frequent fault, thermal-cycle and energy counters can wear storage or corrupt first-fault evidence during abrupt power loss. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is accelerated endurance and power-cut test. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

CALIBRATION LIMITS, VARIANT CODING & CHANGE TRACEABILITY

Detailed subsystem PRD / engineering assurance annex - configuration closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA23-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for calibration limits, variant coding & change traceability; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA23-02	Power rating, grid region, battery range, thermal limits and control maps shall be signed/range-checked and matched to the serialized BOM/hardware revision.	Variant mismatch and calibration-authentication test.
OBCA23-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that A wrong calibration or variant code can create a fully functional but unsafe configuration. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is variant mismatch and calibration-authentication test. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

MANUFACTURING PROCESS CAPABILITY & CRITICAL CHARACTERISTICS

Detailed subsystem PRD / engineering assurance annex - manufacturing closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA24-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for manufacturing process capability & critical characteristics; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA24-02	Critical characteristics, process capability, measurement uncertainty and trim boundaries shall be configuration-controlled; trim shall not conceal structural defects.	Gauge capability, golden-unit and process-correlation evidence.
OBCA24-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Process drift in creepage, TIM thickness, torque, solder quality, shunt value or magnetics can consume design margin while EOL still appears nominal. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is gauge capability, golden-unit and process-correlation evidence. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

EOL HIPOT, FUNCTIONAL POWER TEST & KEY INJECTION

Detailed subsystem PRD / engineering assurance annex - manufacturing closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA25-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for eol hipot, functional power test & key injection; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA25-02	The EOL flow shall define HiPot/IR, low/full-power functional coverage as applicable, gate inhibit, sensing, discharge, network and secure provisioning acceptance criteria.	Qualified EOL fixture with GR&R/golden-unit correlation.
OBCA25-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that EOL must prove insulation, core power functions, protection and unique security identity without overstressing the new unit. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is qualified eol fixture with gr&r/golden-unit correlation. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

FULL-ENVELOPE CHARGE POWER DVP

Detailed subsystem PRD / engineering assurance annex - verification closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA26-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for full-envelope charge power dvp; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA26-02	The DVP shall cover released charge corners, thermal steady-state, transients, credible faults and objective pass/fail criteria fixed before execution.	Configuration-controlled AC/grid emulator and battery emulator test matrix.
OBCA26-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Nominal-room tests can miss combined line, battery, coolant and tolerance conditions that set real charging limits. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is configuration-controlled ac/grid emulator and battery emulator test matrix. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

FULL-ENVELOPE V2G / REVERSE-POWER DVP

Detailed subsystem PRD / engineering assurance annex - verification closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA27-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for full-envelope v2g / reverse-power dvp; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA27-02	The DVP shall cover all released V2G regions, reversals, grid disturbances, communication faults and degraded thermal states.	Bidirectional grid/battery emulator DVP with raw waveform retention.
OBCA27-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Reverse power stresses control authority, grid synchronization, battery discharge limits and some passive elements differently from charging. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is bidirectional grid/battery emulator dvp with raw waveform retention. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

ENVIRONMENT, VIBRATION, INGRESS & SEQUENTIAL STRESS

Detailed subsystem PRD / engineering assurance annex - qualification closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA28-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for environment, vibration, ingress & sequential stress; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA28-02	The qualification plan shall define temperature, vibration, ingress and other vehicle-environment categories from the actual installation and verify post-stress electrical margins.	Sequential environmental + electrical retest.
OBCA28-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Sealing, insulation, solder, connectors and thermal interfaces can degrade only after combined environment sequences. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is sequential environmental + electrical retest. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

MISSION PROFILE, POWER CYCLING & 15-YEAR RELIABILITY

Detailed subsystem PRD / engineering assurance annex - lifecycle closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA29-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for mission profile, power cycling & 15-year reliability; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA29-02	The released mission profile shall drive power-cycling, capacitor, magnetics, connector/seal and NVM life models with documented confidence and margin.	Reliability prediction plus accelerated/mission-profile correlation.
OBCA29-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that A single full-power efficiency point does not prove 15-year life; cycling amplitude and dwell dominate many failure mechanisms. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is reliability prediction plus accelerated/mission-profile correlation. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

SUPPLIER CHANGE, SECOND SOURCE & RELEASE EVIDENCE CLOSURE

Detailed subsystem PRD / engineering assurance annex - release closure for 800 V-class 22 kW bidirectional OBC.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
OBCA30-01	The released product variant shall explicitly configuration-control the physical operating envelope, interfaces, ownership, limits and assumptions for supplier change, second source & release evidence closure; no installation, supplier, calibration or software assumption at this boundary may remain implicit.	Architecture / ICD / configuration review.
OBCA30-02	Evidence reuse after supplier/topology/software changes shall be allowed only inside documented similarity bounds; affected margins and tests shall be reopened otherwise.	Change-impact review and targeted regression DVP.
OBCA30-03	When the qualified margin or required data for this function is unavailable, the system shall enter a non-permissive or derated state that does not increase energy, current, voltage, temperature or protection permissiveness.	Boundary-condition and fault-injection test with physical response recorded.

Verification, margin and future-change rule

The principal engineering concern is that Second-source or technology substitutions can preserve nominal datasheet ratings while changing switching, thermal, insulation or EMC behavior. The design argument shall keep the relevant electrical, thermal, timing, insulation, software and installation assumptions tied to the released hardware/software variant. Primary evidence is change-impact review and targeted regression dvp. Any change that invalidates those assumptions requires impact review and targeted regression before production equivalence is claimed.

REV A1 RELEASE CLOSURE - ENGINEERING ASSURANCE BASELINE

Requirements, evidence, configuration and future-change closure.

The OBC-V2G22 Rev A1 baseline combines the uploaded source-defined product concept with a detailed system PRD and subsystem engineering-assurance annex. Release requires closure of variant-specific TBC values, reconciliation with the vehicle safety/cybersecurity lifecycle and objective evidence for each atomic requirement. No requirement is considered complete only because a related higher-level paragraph exists; completion is tied to the specified test, analysis, inspection, timing trace or configuration evidence.

Gate	Evidence required / release blocker
Architecture / ICD	Released limits, variants, state/authority model and interfaces. Blocker: unbounded or conflicting ownership.
Safety / protection	Closed fault paths, timing/SOA and ISO 26262 work-product traceability. Blocker: open high-severity mechanism or unverified FTTI.
HW / SW	WCCA, control stability, timing/WCET, calibration bounds and configuration integrity. Blocker: unverified derived requirement or margin deficit.
Verification	DVP with calibrated instrumentation, raw evidence and explicit pass/fail. Blocker: failed/waived critical test without approved rationale.
Manufacturing	Critical characteristics, HiPot/EOL capability, programming/provisioning traceability. Blocker: process cannot demonstrate repeatable margin.
Lifecycle	Mission profile, aging model, field counters, service limits and obsolescence controls. Blocker: unsupported 15-year life or change equivalence.

Future-proof rule. Reuse of existing evidence after a design, supplier, vehicle-interface or standards change is allowed only inside documented similarity bounds. Any change affecting electrical stress, control stability, timing, insulation, cybersecurity authority, thermal resistance, reliability or interface ownership shall trigger targeted impact analysis and regression DVP before production equivalence is claimed.

Candidate standards / work products to confirm by program

- Source-defined: ISO/SAE 21434; CAN FD with SecOC; ASIL-C safety intent and hardware overcurrent lockouts.
- Automotive safety/process candidates: ISO 26262 item/HARA/FSC/TSC, ASPICE work products and program-specific OEM requirements.
- Charging/grid candidates for confirmation by market: ISO 15118 / IEC 61851 family and applicable regional V2G/grid-code requirements.
- Vehicle electrical/environment/EMC candidates: applicable ISO 16750 / CISPR 25 / UN ECE R10 / OEM HV and insulation specifications.
- These candidate references are engineering expansion topics, not source-released requirements; edition, scope and applicability must be confirmed by the responsible compliance owner.

Source interpretation boundary

Authoritative source statements are the ten-section baseline in the uploaded 15-product portfolio. All additional numeric acceptance limits not explicitly present in that source remain TBC or controlled parameters until approved by the responsible engineering owner. The structure, atomic-requirement style and evidence-based release philosophy are modeled on the uploaded EMA-FC540 detailed PRD, but aviation-specific standards and requirements are not copied into this automotive product.